CAS (Heimdall) – requirements for OPC UA clients of the PLC
HMI, SiteControl and tools

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Abstract

Every program that talks OPC UA to the CAS PLC shares one small controller with the control program. This document states what such a client must do, and must not do, so that it gets reliable data and does not load or disturb the PLC. It applies to the CAS web HMI (`heimdall-cas-hmi`), to SiteControl services that read or command the PLC, and to engineering tools. The requirements come from the CAS commissioning on the simulator (November 2026), where a client reading every value every second turned out to use about a third of the PLC's CPU. Items not decided are marked OIn and listed in Section 4.

Contents

1 Scope and the server **1**

1.1 Clients in scope . . . 1

1.2 The server . . . 2

1.3 What the measurements showed . . . 2

2 Requirements **2**

2.1 Connection and session . . . 2

2.2 Reading data . . . 3

2.3 Link supervision and data quality . . . 3

2.4 Writing . . . 4

2.5 The contract with the PLC . . . 4

2.6 Engineering tools . . . 5

3 Verification **5**

4 Open issues **5**

1 Scope and the server

1.1 Clients in scope

- **CAS web HMI** (`scaleaq/heimdall-cas-hmi`, Python `asyncua`): reads about 600 values, writes operator commands and alarm acknowledges.

- **SiteControl services** that integrate the CAS PLC (edge drivers on `sitecontrol-opcua-lib`, the OPC Foundation .NET stack): read plant values and alarms, possibly send commands.
- **Engineering tools**: UaExpert, the CODESYS IDE (online monitoring, trace), used during commissioning and service only.

Not in scope: the PLC as an OPC UA *client* (its data sources to the Framo pump controllers), and Modbus clients.

1.2 The server

- WAGO PFC200 G2 (750-8212), **one CPU core**, 512 MB RAM, CODESYS runtime with its OPC UA server on port 4840. The same core runs the control task (20 ms), the I/O, the Modbus masters and the PLC's own OPC UA client.
- The address space is the application's *Symbol Configuration*: `GVLMain`, `GVLIO` and the variables ticked there. Node ids are CODESYS string ids, `ns=4;s=|var|<device>.Application.<GVL>.<path>`; device name and namespace index differ between projects.
- The real PLC requires a user login; anonymous access is off.
- A download, Reset Origin or reboot restarts the address space. An online change can add variables without ending existing sessions.

1.3 What the measurements showed

Simulator PLC 10.1.2.41, 2026-11-03, `htop` on the controller. The HMI read 455 values and 28 quality flags every second and 136 data log values every 2 s, each by string node id, in requests of 50.

State	Runtime main thread	`OPCUAServer` thread	Total CPU
No application loaded	≈15 %	≈5 %	26 %
Application running, HMI blocked	40 %	≈0	33 %
Application running, HMI reading	55–65 %	21–25 %	66–68 %
Application stopped, HMI reading	53 %	25 %	57 %

The cost followed the client, not the program: it stayed with the application stopped, and disappeared when the client was blocked. The server resolves a string node id on every read. A subscription lets it resolve each variable once (`heimdall-cas-hmi` PR #25, to be measured).

2 Requirements

"Shall" is a requirement, "should" a recommendation.

2.1 Connection and session

ID	Requirement	Why / source
CON-01	The client shall log in with its own user account on the PLC. Credentials shall come from the deployment's secrets (environment, Kubernetes secret), never from source code or images.	Real PLC has no anonymous access. One account per client makes a misbehaving client identifiable and lets it be locked out alone.

ID	Requirement	Why / source
CON-02	The client shall use one session per PLC and give it a session name that identifies the client and its site.	Several sessions per client multiply the server load; the name shows up in the runtime log.
CON-03	After a lost connection the client shall reconnect by itself, with a back-off that starts at about 1 s and grows to at most 15 s between attempts.	Fast reconnect after a download; no connection storm while the PLC reboots, and back within 15 s once it is up.
CON-04	On every new session the client shall look up its nodes again, and shall not reuse node ids from an earlier session.	Download, Reset Origin and reboot rebuild the address space.
CON-05	A node that is not found shall be reported once (log, and "not available" where it is shown), shall not stop the client, and shall be looked up again at least every 10 minutes and at every reconnect.	Symbols added by an online change only appeared in the HMI after its pod was restarted.
CON-06	The client shall not hard-code the namespace index or the device name. It should find the GVL root (`GVLMain`) once per session by browsing and derive the other paths from it.	Device and application names differ between the simulator PLC, the site PLC and the soft PLC.
CON-07	Security policy and certificates OI1.	Not decided for site use.

2.2 Reading data

ID	Requirement	Why / source
RD-01	Cyclic data shall be read through a **subscription** (monitored items), not by reading every value every cycle.	Section 1.3: polling 600 values per second used about a third of the CPU.
RD-02	The sampling and publishing interval shall be at least 1 s. Faster sampling shall be limited to a few values, only while a user looks at them, and at most 5 Hz.	The HMI's wave trend samples two level transmitters at 5 Hz only while a page shows it.
RD-03	Monitored items shall use queue size 1 (latest value only).	The client needs the current state, not a history; queues cost server memory.
RD-04	A client that must read (no subscription available, or a value the server refuses to monitor) shall read at most 50 values per request and should not read the same value more often than every 2 s.	The PFC200 refuses large requests (`BadTooManyOperations` at 200).
RD-05	Large browses (for example all `AlarmAck` bits, about 1100 nodes, 35 s) shall run at most once per session, in the background, and never periodically.	Browsing walks the symbol tree in the runtime.
RD-06	The client shall read only what it uses. Each client shall keep its list of paths in one place in its code, with the PLC data type.	The list is the contract with the PLC (Section 2.5).
RD-07	Total load: with all permanent clients connected the PLC shall keep TBD CPU in reserve OI2.	Measured baseline 33 % without clients.

2.3 Link supervision and data quality

ID	Requirement	Why / source
LQ-01	The client shall notice a lost connection within 3 publishing intervals (keep-alive, or a cheap read of `Server_ServerStatus_CurrentTime`, a numeric node).	A subscription alone sends nothing when nothing changes.
LQ-02	While the connection is lost, every value shall be shown as unknown, never as the last value received.	An operator must not act on frozen data.
LQ-03	The client shall respect the PLC's own quality information: values whose quality flag says bad (`Status.BadQuality`), and values from a device whose link the PLC reports down or stale (`CommOK` FALSE, `DataStale` TRUE), shall be shown as invalid.	The PLC knows more about its field devices than the client.
LQ-04	The client shall show what the PLC computes (scaling, averages, states) and shall not compute plant values itself.	One source of truth; the FDs describe the PLC.
LQ-05	Timestamps: the client shall use its own clock for when it received a value until the PLC clock is synchronised **OI3**.	The simulator PLC clock was months off.

2.4 Writing

ID	Requirement	Why / source
WR-01	The client shall write only variables on its own whitelist: `Command` members, `AlarmAck` bits and named setpoints. It shall never write `Status`, `Input`, `Output`, `InternalCommand` or `Alarm`.	The PLC owns its state.
WR-02	Commands are pulses: write TRUE, hold at least 1 s or until the PLC clears it, then FALSE. The client shall not rely on the PLC clearing a bit it did not document as cleared.	AlarmManagement acknowledges on the rising edge; some commands are latched and cleared by the PLC.
WR-03	On connect and on a clean session end, the client shall write FALSE to every command bit it may have left TRUE.	A command bit left TRUE after a lost link would act again on the next edge.
WR-04	Setpoints shall be bounds-checked by the client before writing, with the same limits as the PLC.	The PLC limits them too; the client gives the operator a clear message instead.
WR-05	Retained configuration (`InletConfig`, `OxyConfig`, IP addresses) shall only be written by explicitly named, whitelisted functions, never in bulk.	Retained values survive a restart; a wrong bulk write survives too.
WR-06	Every client shall have a read-only mode that writes nothing.	Diagnosis on a running plant.
WR-07	Only one client shall command a given function at a time **OI4**.	HMI and SiteControl must not fight over the same command.

2.5 The contract with the PLC

ID	Requirement	Why / source
CT-01	Everything a client reads or writes shall be in the Symbol Configuration. New fields of a structure shall be ticked there explicitly (where the structure is not exported whole).	New fields were in the PLC but not on the server until ticked.
CT-02	A PLC change that renames, removes or retypes a published variable shall update the clients' path lists in the same change, or be agreed with the client owners first.	The client finds out only at runtime.
CT-03	The PLC should publish a program version (`PlcInfo`) that clients log at connect.	To tell which program a client talked to.

2.6 Engineering tools

ID	Requirement	Why / source
EN-01	UaExpert, the CODESYS IDE online view and traces shall be connected only while needed, and closed after commissioning and service.	They add load (subscriptions, monitoring) on the same core.
EN-02	CPU checks on the controller shall be done with the IDE logged out and UaExpert closed, reading `htop` with thread names (`OPCUAServer`, `codesys3`).	Otherwise the tools measure themselves.

3 Verification

A client is accepted when, on the simulator PLC with the application running:

1. total CPU with the client connected is within TBD of the value with the client blocked (`iptables -I INPUT -p tcp --dport 4840 -s <clientIP> -j REJECT`, removed afterwards) **OI2**;
2. it notices a lost link within 3 publishing intervals and shows its values as unknown;
3. it reconnects by itself after a download and after a reboot, and finds its nodes again;
4. it finds a variable added by an online change without a restart (CON-05);
5. it writes nothing in read-only mode.

4 Open issues

No.	Issue	To decide
OI1	Security policy for site use: None, or Basic256Sha256 with certificates; who manages them.	Automation / IT.
OI2	CPU budget: how much the PLC shall keep in reserve with all clients connected, and how much one client may use.	After the subscription has been measured (PR #25).
OI3	NTP on the PLC; until then source timestamps are not trusted.	Commissioning.
OI4	Which client commands what when both the HMI and SiteControl are connected.	Project.

No.	Issue	To decide
OI5	Does SiteControl read the CAS PLC directly over OPC UA, or through the HMI or another gateway?	SiteControl team.